New Zealand Journal of Educational Studies (2025) 60:435–449
https://doi.org/10.1007/s40841-025-00405-2

ARTICLE

The Impact of Better Start Literacy Approach Teaching on the Retention of Children's Early Literacy Skills over the Summer Holidays

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Received: 19 December 2024 / Accepted: 19 August 2025 / Published online: 24 October 2025
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Abstract
Studies have documented a Summer Learning Effect (SLE), where children lose some of their school learning over the summer holiday period. Few of these studies, however, have investigated the impact of classroom teaching on the SLE. This study investigated the SLE on five and six-year-old emergent learners in New Zealand who had 10 weeks of early literacy instruction using the Better Start Literacy Approach (BSLA). Results indicated no statistically significant change in student scores over the summer holiday break. These results indicate that the BSLA may be effective in embedding initial literacy skills.

Keywords Summer learning effect · Structured literacy · Emergent literacy · New Zealand

Introduction

We're all goin' on a summer holiday
No more workin' for a week or two
Fun and laughter on our summer holiday
No more worries for me or you
For a week or two
(Welch & Bennett, 1963).

The joy of summer holidays is as real now as it was for Cliff Richard and The Shadows in 1963. School children all over the world look forward to 'fun and laughter',

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not just for a week or two, but for anywhere between 6 and 12 weeks, depending on which part of the world they are from. While children may look forward to 'no worries' during this time, many teachers and parents are concerned about the effect of summer holidays on children's learning. This paper examines the effect of the summer holiday break on the literacy learning of a group of five-year-old children in their first year at school in New Zealand. Specifically, it addresses the question:

> Have the foundational literacy skills (e.g. phoneme awareness, spelling, and word reading) children gained after they have received 40 lessons of the Better Start Literacy approach been sustained over the summer holiday break?

The Summer Learning Effect

The Summer Learning Effect (SLE) has been documented internationally and is the phenomenon whereby some children start the new school year with achievement levels lower than at the end of the previous school year. This is referred to as either a slide, learning loss or setback (Meyer et al., 2019). These losses can be quite substantial. In a 1996 meta-analysis of 39 studies, Cooper et al. (1996) concluded that "summer loss equalled about one month on a grade-level equivalent scale" (p. 227). Of particular interest in these studies is the SLE for children from low socio-economic backgrounds. Economically disadvantaged children in the USA made good progress in literacy during the school year. During the summer break, however, they regressed in their learning due to the lack of out-of-school resources. Similarly, Coley et al. (2020) found that parental socioeconomic status (SES) disparities impacted children's retention of learning over the summer period. Children from higher SES homes made greater gains in reading over the summer months than those from lower SES homes. Also of interest is Cooper et al.'s (1996) suggestion that studies measuring the retention of factual and procedural skills (e.g. maths and spelling skills) showed larger losses than those measuring conceptual skills (e.g. maths problem solving and reading comprehension).

Outside of the United States, Slade et al. (2017) conducted a study in Malawi where the summer break is approximately seven weeks long. Students transitioning from grades 1–2 and 2–3 had significant grade-transition literacy losses over the summer. In a study conducted in Germany, Meyer et al. (2017) tested whether German students, who have a 6–10-week summer break (rather than the 10–12 weeks of US students), also show a summer learning effect. They investigated whether second grade students in high and low-income schools had differing literacy results at the end of their summer break and found there was a loss in reading comprehension and writing over the summer period, with children from low-income schools showing the greatest loss. These losses were evident for both reading comprehension and writing. Thus, even with a shorter summer holiday break, literacy losses were similar to those reported in North America.

Studies into SLE in North America, Europe and Africa focus mainly on SLE learning losses of those at different SES levels, or on the effect of summer literacy pro-

grammes (Albee et al., 2019; Graham et al., 2011). The pattern is also evident in New Zealand studies. Similar to Meyer et al. (2017), Tiruchittampalam et al. (2018) analysed, among other things, whether schools in higher or lower SES areas in New Zealand were able to narrow the literacy achievement gap over the summer. The New Zealand summer school break begins mid-December and ends in early February - approximately six weeks in total. Students in the study were five-year-olds who began school in the middle of the year. Children were tested at school entry, at the end of the year (December), and at the beginning of the following year (February). The average gap between assessment points over the summer was 2.4 months. They found that children attending higher SES schools usually had gains over the summer, whereas the opposite was true for children attending lower SES schools.

Investigating the effect of SES levels on the SLE in writing, Meyer et al. (2019) looked at the writing results of students in Years 4–7 (8–12-year-olds) in New Zealand. They found that a considerable loss of learning was sustained by students, and particularly by ethnic minority students in lower SES schools. Williamson et al. (2020) researched the effects of a summer blogging programme on writing and reading outcomes. They found that Year 5–10 students in several low-income New Zealand communities could benefit from posting regularly to a blog over summer. However, these effects were not universal.

Several New Zealand studies have investigated the effects of reading programmes over the summer (e.g., Meyer et al., 2019; Nicholson & Tiru, 2019; Turner & Tse, 2015). Nicholson and Tiru trialled the effectiveness of a summer reading intervention for 36 low SES children aged 5 years 6 months to 10 years 5 months. They compared their results to 36 matched controls. Findings from their three-week intervention during the summer holidays showed gains in reading accuracy and basic decoding skills, but not reading comprehension. In addition, one South Auckland primary school planned and implemented a summer reading programme in collaboration with their local library (Cromie et al., 2017). Their aim was to improve reading achievement levels of Year 5 students in four classes by increasing reading engagement during the summer holidays. Prior to the summer project, the regular class reading programme was complemented by enhanced literacy sessions taken by a research-support teacher who took groups of target students. Findings from their inquiry suggest that the 'slide' in reading achievement can be minimised, but a summer reading programme alone is insufficient for long-term gains in reading. Other factors that helped to minimise the slide were teacher awareness of the impact of their practices on motivation and engagement, the significance of developing a 'community of readers' and teachers reading novels aloud to their year-group classes.

Challenges with Current Literacy Progress in New Zealand

Understanding children's retention of early literacy skills over summer is important within the larger challenge of addressing inequities within New Zealand. An indication of these inequalities is evident when New Zealand literacy rates are compared to

similar countries. Literacy results in New Zealand have been falling in comparison to many other OECD countries since 2011. Progress in International Reading Literacy Study (PIRLS) data revealed a statistically significant drop in ten-year-old New Zealand children's mean reading scores between 2011 and 2016. In the 2016 report New Zealand scored 24th out of the 26 OECD nations who participated in PIRLS (Mullis et al., 2017).

In a 2013 article describing reasons for falling literacy rates in New Zealand, Tunmer et al. suggest poor pedagogy. They conclude that the predominantly constructivist approach to literacy, rather than routine teaching of literacy skills, has led to poor results. Tunmer et al. describe this constructivist teaching as the whole language approach where there is very little explicit or structured teaching of foundational literacy skills such as phonemic awareness or the alphabetic code. Tunmer, Chapman and others have repeatedly called for a change to literacy instruction in New Zealand (Chapman et al., 2018; Hood & Taylor, 2022; Hughson & Hood, 2022; Prochnow et al., 2015; Tunmer et al., 2013; Tunmer et al., 2015). Chapman et al. (2018) suggest that classroom teaching should incorporate systematic phonics instruction that includes training in effective word-level decoding. Hood and Taylor (2022) echo this sentiment, stating that explicit phonics instruction, along with the building of vocabulary through engaging with rich language, is essential in primary education. Crucially, this needs to be done in culturally responsive and sustaining ways which make use of culturally-relevant learning contexts and pedagogical approaches (Suggate, 2016).

Limitations in Previous Studies

The studies on the SLE have been conducted in various countries - many of which have summer breaks that vary in duration (e.g. New Zealand has a six-week break while the USA has up to three months). Additionally, most studies focus on disparities in children from differing SES and/or how summer learning programmes help ease learning losses experienced by children from low SES families.

A gap in these studies is the effect of ordinary classroom instruction on the SLE. Very few studies investigate whether the type of literacy instruction children receive during the school year has an impact on their ability to retain literacy skills over the summer holiday break. Additionally, few studies on the SLE have been conducted in New Zealand. This study seeks to address these gaps by investigating the following question:

> Are the foundational literacy skills that children acquire in response to the implementation of the Better Start Literacy Approach (i.e. phoneme awareness, spelling and word decoding skills) sustained over the 6–8 week summer holiday period?

Brief Description of BSLA

The Better Start Literacy Approach (BSLA) (Gillon et al., 2024) emerged from controlled research trials funded through the Better Start National Science Challenge in New Zealand—(Gillon et al., 2019, 2020). Findings from these trials indicated that children who received BSLA teaching showed accelerated growth in their non-word reading, vocabulary and phoneme awareness skills compared to children in classrooms who received other types of early literacy teaching. Subsequent to the success of the controlled research trials, the Ministry of Education contracted the lead researchers of BSLA to implement large scale professional learning and development for teachers across the country to implement BSLA.

The BSLA is an example of a structured literacy teaching approach based on the science of reading. It was developed specifically for the New Zealand cultural context and adopts a culturally responsive and inclusive teaching framework. Using explicit and structured teaching techniques, teachers develop children's foundation literacy skills in four key areas: (1) Phonological and morphological awareness; (2) Oral language (vocabulary, oral narrative and listening comprehension skills); (3) Transferring phoneme awareness skills to reading and spelling; and (4) Reading connected text using decodable texts (the Ready to Read Phonics Plus readers) (Arrow et al., 2021).

Online and paper and pencil assessment tasks were specifically developed to monitor children's literacy growth in response to BSLA teaching during their first year of school and to provide specific direction for teaching to address individual needs (Gillon et al., 2024; Scott et al., 2023). These tasks are described in detail in Scott et al. (2023) and assess a child's ability to:

- Identify an initial phoneme (using a 10-item test). Cronbach's alpha across the 10 items of 0.85, indicating high reliability (Gillon et al., 2024).
- Blend phonemes (using a 10-item test). Cronbach's alpha across the 10 items of 0.79, indicating good reliability (Gillon et al., 2024).
- Match sounds to letters (using a 10-item test with 8 items in Set 1 and 17 in Set 2). Cronbach's alpha across the 10 items was 0.84 for Set 1 and 0.83 for Set 2, indicating high reliability (Gillon et al., 2024).
- Read non-words. There are four sets of non-word reading tests which each contain 10 words and increase in difficulty. The two-way random effects intraclass correlation is 0.92 across the 10 items across sets, indicating high inter-rater reliability (Gillon et al., 2024).
- Spell non-words. The same non-words that are used in the non-word reading assessment are used in non-word spelling. The two-way random effects intraclass correlation was 0.93 across the 10 items across sets, indicating high inter-rater reliability (Gillon et al., 2024).

The proficiency scores for each of these tasks is included in Table 1 below.

Table 1 Overview of assessment tasks and associated scores

Assessment Task	Maximum Score	Proficiency Score (= or >)
Initial Phoneme Identity	10	8
Letter Sound Knowledge (Set 1)	8	6
Letter Sound Knowledge (Set 2)	17	13
Phoneme Blending	10	8
Non-Word Reading (Set 1)	60	48
Non-Word Reading (Set 2)	72	57
Non-Word Reading (Set 3)	66	53
Non-Word Reading (Set 4)	66	53
Non-Word Spelling (Set 1)	60	48
Non-Word Spelling (Set 2)	72	57
Non-Word Spelling (Set 3)	66	53
Non-Word Spelling (Set 4)	66	53

Current SLE Study

This study investigated the SLE on five and six-year-old emergent learners in New Zealand who had 10 weeks (40 lessons) of early literacy instruction using the Better Start Literacy Approach (BSLA). Children's ten-week assessment results at the end of the year were compared with their scores on those same assessments at the beginning of the following year. These assessment points were within the final 45 days of the school year and within the first 15 days of the following year after the New Zealand summer holiday break of approximately six weeks. Children's ten-week assessment results at the end of the year were compared with their scores on those same assessments at the beginning of the following year.

Method

Participants

Informed consent was gained from principals and parents, and assent was gained from children before the study began, after full ethics approval had been granted by the University of Canterbury Human Ethics Research Committee (HREC 2022/134).

Schools that had been a part of the BSLA micro-credential training were invited to participate. Children were then eligible to take part in the research if they:

- had a baseline assessment and began BSLA instruction in the third term of the school year.
- had received at least 40 BSLA sessions by the fourth (i.e. last) term of the New Zealand school year and had completed a 10-week assessment.

As participants were five or six-year-old children who were not able to give informed consent for access to initial baseline data, this was sought from parents via a Qualtrics survey.

Data Collection

The assessments were administered three times (see Fig. 1):

1. at school entry and prior to any BSLA teaching. This data was collected by teachers to inform next steps for learning and offered an indication of baseline strengths. They were administered by teachers, facilitators or teacher aides trained in the BSLA.
2. at 10 weeks (after 40 BSLA sessions). These were again administered by teachers, facilitators or teacher aides as part of the end of year school assessment schedule which is conducted in the final 45 days of the school year.
3. after the summer holiday in the first 15 days of the school year. BSLA facilitators, BSLA lead facilitators or UC BSLA staff re-administered these tasks, as BSLA classroom teachers are not usually expected to re-assess students at this point.

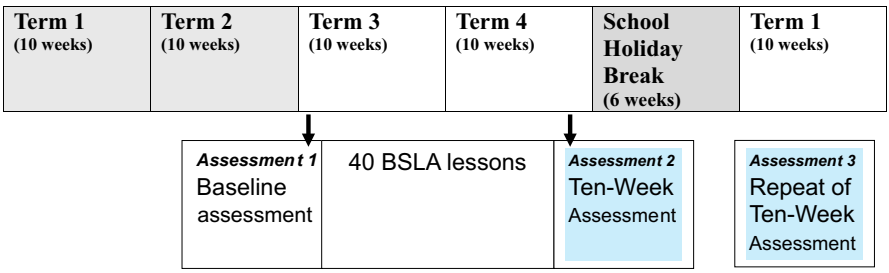


Fig. 1 Assessment timepoints

Relevant BSLA assessment tasks were used to assess children's progress. The range of assessment tasks and maximum scores are summarised in Table 1. At baseline, if proficiency (80% or above) was reached in phonological awareness tasks (Initial Phoneme Identity, Letter Sound Knowledge Set 2 and Phoneme Blending), children were assessed on Non-Word Reading. Children progressively worked through the Non-Word Reading tasks until they no longer reached proficiency (80% or above). Non-Word Spelling was administered at the level that corresponded to the highest level reached in Non-Word Reading. Sets of tasks increase in difficulty level. Letter Sound Knowledge Set 2 introduces more complex phonemes (e.g. 'ch') for identification. Similarly, Non-Word Reading and Non-Word Spelling tasks progress in difficulty from reading and spelling single letters to recognising and spelling split digraphs, r-controlled vowels and complex morphemes.

Identical assessments were administered at timepoints two and three. Scores at timepoints two and three were then compared to check which skills, if any, were retained over the six-week summer holiday break.

Prior to the assessments being re-administered after the summer break, all assessors were given a script to read out, and a form for students to indicate assent or otherwise by colouring in either a smiley or sad face.

Results

Demographics

Fifty-four students were assessed at the designated time points. Six students were subsequently excluded as they did not meet the criteria for having 10 weeks of BSLA teaching and assessment at both time points.

Students were assessed from existing BSLA schools which were sampled from around the country with two schools from Wellington, three from the Canterbury area and two from Auckland. Schools in New Zealand are divided into decile rankings with Decile 1 schools representing the 10 per cent of schools with the highest proportion of students from low socio-economic communities. Decile 10 schools are the 10 per cent of schools with the lowest proportion of students from these communities. In this study, 6 students were from Decile 8–10 schools, 43 students were from Decile 5–7 schools and 5 were from Decile 1–4 schools. Thus, the majority of students were from mid socio-economic communities. The average age at the third assessment timepoint was 6 years 0 months. Children's ethnicities were New Zealand European (62%), Asian (33%) and Other (5%), with 1 per cent unknown.

Data Analysis

Python was used to run statistical tests on the dataset. Pair-wise T-tests were used to determine if there were significant differences between the before and after assessment scores for each assessment task. Cohen's D was also calculated for each measure to determine the effect size between the second and third time points. A value of ~0.2 indicates no/small effect, ~0.5 a medium effect and ~0.8 a large effect. In

addition, a Reliable Change Index (RCI) was calculated. A change index is calculated using the ratio of score improvement and the standard error of the difference. An RCI of greater than 1.96 implies a statistically significant difference. Further analysis based on individual student scores was carried out to see if there were any trends.

For this study, we calculated a score difference variable to classify students to one of two categories: improved/retained or regressed for each measure. This variable was included to account for small increases or decreases that may be due to measurement error, rather than true change. For the initial phoneme identity, phoneme blending, and letter-sound knowledge assessment tasks, which had few items, a decrease in score of greater than 1 would be considered a regression in skill. For Non-Word Reading and Non-Word Spelling tasks, which had a score range of 60–72, a decrease in score of greater than 2, would be considered a regression in skill. All other score variations would be considered to have improved/retained. For BSLA assessments, score variation has not been determined for students for the same skill. Due to the number of students who were recorded doing each assessment, analysis of the breakdown of student achievement in terms of ethnicity, school decile, school size or location would not yield enough students in the dataset to draw confident conclusions.

Quantitative Results

A summary of the results and number of students who attempted each task is shown in Table 2. Because of the low number of students who attempted Non-Word Reading and Non-Word Spelling tasks at Set 4, ($n=4$ and $n=1$, respectively) these were excluded from subsequent analyses. The number of students, therefore, relates to those who did the task at both time points.

Figures 2 and 3 show the number of measures for which each student achieved either a higher or a lower score and the number of reported measures in which each student was assessed.

Table 2 Score results pre- and post-summer

Measure	Max score	Number of students	Pre-summer mean score (S.D.)	Post-summer mean score (S.D.)
Initial phoneme identity	10	16	8.8 (1.7)	8.5 (1.5)
Letter sound knowledge (Set 1)	8	17	7.2 (1.1)	7.3 (1.5)
Letter sound knowledge (Set 2)	17	17	15.1 (2.3)	14.2 (1.8)
Phoneme blending	10	21	8.5 (1.5)	8.8 (2.0)
Non-word reading (Set 1)	60	38	40.4 (17.5)	39.3 (16.4)
Non-word reading (Set 2)	72	19	61.6 (5.7)	61.3 (5.5)
Non-word reading (Set 3)	66	14	49.9 (6.6)	50.8 (6.9)
Non-word reading (Set 4)	66	4	46.8 (2.2)	45.1 (3.8)
Non-word spelling (Set 1)	60	40	47.0 (13.5)	45.3 (14.7)
Non-word spelling (Set 2)	72	26	61.8 (6.1)	59 (9.1)
Non-word spelling (Set 3)	66	20	47.1 (5.5)	50 (5.3)
Non-word spelling (Set 4)	66	1	46.0	39.0[a]

[a]Standard deviation is not shown for this task as only one student attempted this task

The majority of students (34) both retained or improved their overall scores but regressed in at least one measure. Seven students were considered to have retained or improved, and three students to have regressed, in all measures assessed.

Initially, a t-test was used to see if there were significant differences between the two distributions for each task. Using alpha of 0.05, a null hypothesis stated that there was not a significant difference between the Pre-Summer Scores and Post-Summer Scores. The null hypothesis failed to be rejected for all tasks except Non-Word Spelling (Set 2) ($p = .02$).

To determine the effect size between the two timepoints Cohen's D was calculated for each measure. A value of ~0.2 would indicate a small effect, ~0.5 a medium effect, and ~0.8 a large effect.

Additionally, an RCI was calculated for these measures. An RCI of greater than 1.96 implies a statistically significant difference with positive values showing improvement and negative values showing regression. The values are shown in Table 3. Looking at RCI, no measure shows a statistically significant difference. With Cohen's D, most measures show between small-medium effects with only Non-word Spelling (Set 3) showing a medium positive improvement in scores.

The values for each assessment type are listed in Table 3.

For context, the mean score difference is also shown in Table 3. This was calculated for each measure by looking at the score difference for each student and calculating the mean value.

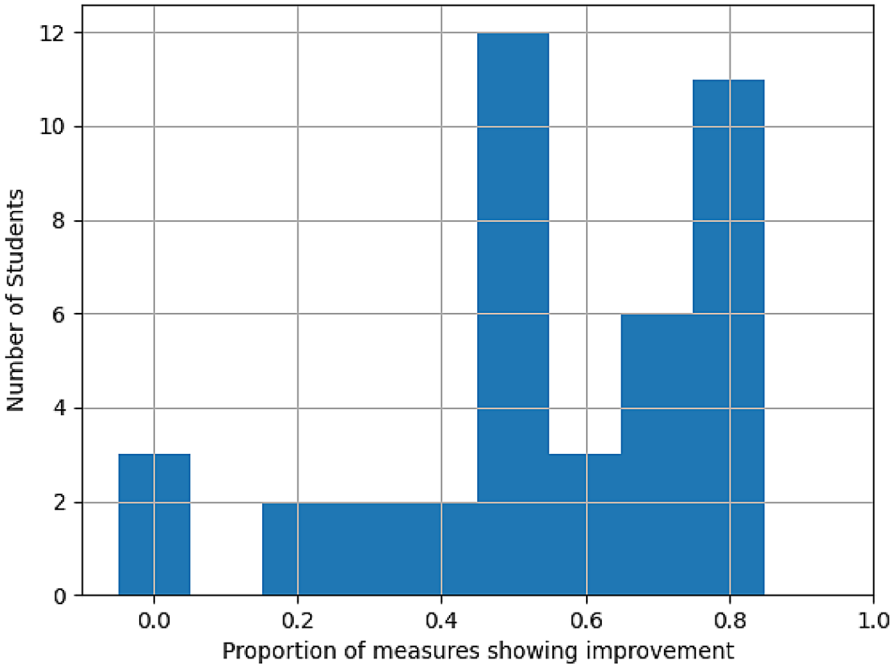


Fig. 2 Proportion of measures where improvement was evident

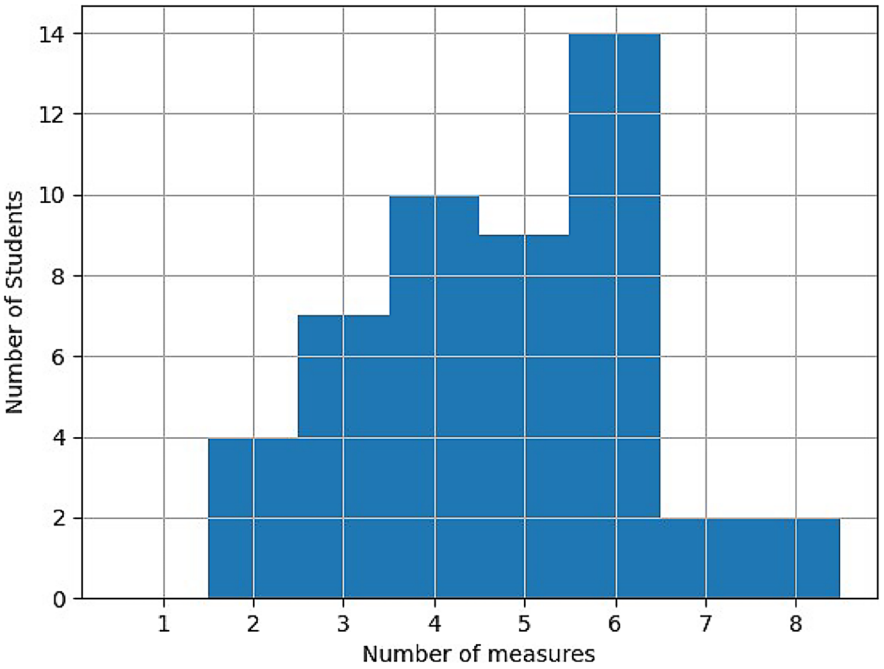


Fig. 3 Number of measures in which students were assessed

Table 3 Cohen's D showing effect sizes, reliable change index (RCI) values and mean of the score difference between pre-summer scores and post-summer scores for each assessment type

Measure	Cohen's D	RCI	Mean Score Difference
Initial Phoneme Identity	-0.16	-0.66	-0.25
Letter Sound Knowledge (Set 1)	0.04	0.14	0.06
Letter Sound Knowledge (Set 2)	-0.41	-1.16	-0.82
Phoneme Blending	0.13	0.46	0.24
Non-Word Reading (Set 1)	-0.07	-0.65	-1.11
Non-Word Reading (Set 2)	-0.07	-0.17	-0.37
Non-Word Reading (Set 3)	0.13	0.33	0.86
Non-Word Spelling (Set 1)	-0.12	-1.34	-1.70
Non-Word Spelling (Set 2)	-0.34	-1.75	-2.85
Non-Word Spelling (Set 3)	0.53	1.44	2.80

Discussion

This study investigated whether children who had received 40 lessons using a structured literacy approach, the Better Start Literacy Approach, sustained their literacy learning over the summer holiday break. The sample of forty-nine children came from three distinct areas in New Zealand and represented a range of decile levels. Measures of phonological awareness, letter-sound knowledge, and nonword reading and spelling were conducted at year end and children were reassessed on these same measures immediately after the six-week summer holiday break.

The results indicate that children from a range of school deciles (2–7), who engaged in 40 BSLA lessons, appeared to be less impacted by the summer break compared with findings from existing studies. Assessment of the effect size between the second and third timepoints showed a medium positive improvement in Non-Word Spelling (Set 3) scores and small to medium effects for most of the other measures. Although no measure reached statistical significance, the majority of students (34) both retained or improved their overall scores but regressed in at least one measure. Seven students retained or improved in all measures assessed.

Overall, there was no evidence of a statistically significant change in students' literacy levels over the summer. Small variations in scores before and after the summer holiday break would more likely reflect measurement error than true change. If anything, the scores trend towards improvement rather than regression.

In comparison to previous research, the data in this research is encouraging. There are indications that children have retained their literacy skills over the summer holiday period. This may be due to a number of factors.

Pedagogically, the BSLA incorporates systematic and explicit teaching of phonological awareness, vocabulary instruction, oral language and listening comprehension, and then links these skills to small group reading. Instruction in these four research-validated elements may result in well-embedded skills that are not lost over the holiday break.

In a follow-up study that tested the efficacy of a phonological awareness intervention for 5-7-year-old New Zealand children with spoken impairment, Gillon (2002) found that gains made by children with spoken language impairment who received phonological awareness intervention were sustained for approximately 11 months. In contrast, the control group of children with spoken language impairment who did not receive phonological awareness intervention showed remarkably little improvement in phoneme awareness over time and the majority remained poor readers. This study complements Gillon's earlier work by showing that phonological awareness skills taught through the BSLA lesson plans are also sustained over the summer holiday break by emergent literacy learners.

Another reason for the sustaining of skills over the holiday break may be the consistency of research-validated practice by teachers. Several studies (e.g. Arrow et al., 2019; Stark et al., 2016) have highlighted the importance of teachers' linguistic knowledge and their ability to apply that knowledge to the teaching of beginning readers. When comparing instructional practice to teacher self-reports of their ability to explicitly teach phonological awareness, both Arrow and Stark noted a gap, with teachers being unaware of their poor teaching practice. This suggests that teachers'

linguistic knowledge needs to be accompanied by professional development in the application of that knowledge to classroom practice to ensure effective explicit reading instruction to beginning readers.

The BSLA uses layers of professional assistance to ensure teachers are well-trained in both the theory and the practice of sound literacy instruction. The University of Canterbury's BSLA microcredential course includes 10 weeks of compulsory Zoom teaching. This Zoom teaching covers both the theory and the practice of early literacy instruction, using recent research-validated data. In addition, each teacher is assigned a trained facilitator who works with the teacher at the classroom level to ensure fidelity to best early literacy practice. The success of this approach for classroom teachers has been shown in consistently high student results. A recent report details accelerated progress in almost 30,000 BSLA children's phonic and phonemic awareness skills after only 10 weeks of BSLA teaching (Gillon et al., 2024). This evidence of excellent teacher practice may well contribute to the retention of BSLA children's literacy skills over the summer holiday break.

These results contrast with the findings of previous international and New Zealand SLE studies, where literacy skills were often lost over the summer holiday break, (e.g., Kuhfeld & Soland, 2021; Meyer et al., 2019; Nicholson & Tiru, 2019). This study also makes a unique contribution as it focused on the impact of a particular type of classroom literacy instruction prior to the summer break rather than an additional intervention during the break. Very few studies have investigated whether the type of classroom literacy instruction children receive during the school year has an impact on their ability to retain literacy skills over the summer holiday break. Additionally, few studies on the SLE have been conducted in New Zealand.

Limitations

The results from this study are positive, but preliminary. They therefore need to be interpreted in the light of the following limitations. The sample size in this study was small. A further investigation with a larger sample size and a robust research design would help to confirm the results of this study. There was no comparison group of students where teachers were doing 'business as usual' literacy instruction. The inclusion of a control group in future studies would help validate these findings. Finally, there was no measure of students' exposure to literacy activities during the summer break. Including this data would help to highlight whether the method and content of classroom instruction was the main factor in the sustaining of literacy learning over the summer break.

Conclusion

The results of this study, showing the sustaining of literacy learning over the summer holiday break for a group of students who had received only 10 weeks of BSLA literacy instruction are encouraging. With year-round teaching using explicit, research-based literacy instruction, maybe summer holidays really will be worry-free for all concerned – students, teachers and parents.

Acknowledgements The authors would like to thank the children and teachers who participated in this study.

Funding Open Access funding enabled and organized by CAUL and its Member Institutions

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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References

Albee, J. J., Smith, M. L., Arnold, J. M., & Dennis, L. B. (2019). Digging struggling students out of the summer reading slump. *Reading Teacher, 73*(3), 291–299. https://doi.org/10.1002/trtr.1847

Arrow, A. W., Braid, C., & Chapman, J. W. (2019). Explicit linguistic knowledge is necessary, but not sufficient, for the provision of explicit early literacy instruction. *Annals of Dyslexia*, *69*(1), 99–113. https://doi.org/10.1007/s11881-018-00168-0

Arrow, A. W., Gillon, G., McNeill, B. C., & Scott, A. (2021). *Early literacy initiative: Ready to read phonics plus project*. Child Well-being Research Institute.

Chapman, J. W., Greaney, K. T., Arrow, A. W., & Tunmer, W. E. (2018). Teachers' use of phonics, knowledge of language constructs, and preferred word identification prompts in relation to beginning readers. *Australian Journal of Learning Difficulties, 23*(1), 87–104. https://doi.org/10.1080/19404158.2018.1467937

Coley, R. L., Kruzik, C., & Votruba-Drzal, E. (2020). Do family investments explain growing socioeconomic disparities in children's reading, math, and science achievement during school versus summer months? *Journal of Educational Psychology, 112*(6), 1183–1196. https://doi.org/10.1037/edu0000427

Cooper, H., Nye, B., Charlton, K., Lindsay, J., & Greathouse, S. (1996). The effects of summer vacation on achievement test scores: A narrative and meta-analytic review. *Review of Educational Research, 66*(3), 227–268. https://doi.org/10.3102/00346543066003227

Cromie, R., Fa'alogo, I. G., Gwilliam, M., & Limbrick, L. (2017). A summer reading programme. *Set: Research Information for Teachers (Wellington)*. https://doi.org/10.18296/set.0067

Gillon, G. (2002). Follow-up study investigating the benefits of phonological awareness intervention for children with spoken language impairment. *International Journal of Language & Communication Disorders, 37*(4), 381–400. https://doi.org/10.1080/1368282021000007776

Gillon, G., McNeill, B., Denston, A., Scott, A., & Macfarlane, A. (2020). Evidence-based class literacy instruction for children with speech and language difficulties. *Topics in Language Disorders, 40*(4), 357–374. https://doi.org/10.1097/tld.0000000000000233

Gillon, G., McNeill, B., Scott, A., Denston, A., Wilson, L., Carson, K., & Macfarlane, A. H. (2019). A better start to literacy learning: Findings from a teacher-implemented intervention in children's first year at school. *Reading & Writing, 32*(8), 1989–2012. https://doi.org/10.1007/s11145-018-9933-7

Gillon, G., McNeill, B., Scott, A., Gath, M., Macfarlane, A., & Taleni, T. (2024). Large scale implementation of effective early literacy instruction. *Frontiers in Education*. https://doi.org/10.3389/feduc.2024.1354182

Graham, A., McNamara, J. K., & Van Lankveld, J. (2011). Closing the summer learning gap for vulnerable learners: An exploratory study of a summer literacy programme for kindergarten children at-risk for reading difficulties. *Early Child Development and Care, 181*(5), 575–585. https://doi.org/10.1080/03004431003646525

Hood, N., & Taylor, H. (2022). *Now I don't know my ABC: The perilous state of literacy in Aotearoa New Zealand*. The Education Hub.

Hughson, T., & Hood, N. (2022). What's happening with literacy in Aotearoa New Zealand? Building a comprehensive national picture. The Education Hub. Retrieved from https://theeducationhub.org.nz/wp-content/uploads/2022/03/Ed-Hub_Long-literacy-report_v2.pdf

Kuhfeld, M., & Soland, J. (2021). The learning curve: Revisiting the assumption of linear growth during the school year. *Journal of Research on Educational Effectiveness*, *14*(1), 143–171. https://doi.org/10.1080/19345747.2020.1839990

Meyer, F., Meissel, K., & McNaughton, S. (2017). Patterns of literacy learning in German primary schools over the summer and the influence of home literacy practices. *Journal of Research in Reading*, *40*(3), 233–253. https://doi.org/10.1111/1467-9817.12061

Meyer, F., Yao, E. S., & Meissel, K. (2019). The summer learning effect in writing in New Zealand. *Reading & Writing, 33*(5), 1183–1210. https://doi.org/10.1007/s11145-019-10003-6

Mullis, I. V. S., Martin, M. O., Foy, P., & Hooper, M. (2017). PIRLS 2016 international results in reading. Retrieved from http://timssandpirls.bc.edu/pirls2016/international-results/

Nicholson, T., & Tiru, S. (2019). Preventing a summer slide in reading - The effects of a summer school. *Australian Journal of Learning Difficulties, 24*(2), 109–130. https://doi.org/10.1080/19404158.2019.1635499

Prochnow, J. E., Tunmer, W. E., & Greaney, K. T. (2015). Findings from New Zealand's participation in the progress in international reading literacy study in 2001, 2006, and 2011. In W. E. Tunmer & J. W. Chapman (Eds.), *Excellence and Equity in Literacy Education: The Case of New Zealand* (pp. 23–40). https://doi.org/10.1057/9781137415578_2

Scott, A., Gillon, G. T., & McNeill, B. (2023). *An innovative approach to early literacy assessment through the Better Start Literacy Approach*. Literacy Forum NZ.

Slade, T. S., Piper, B., Kaunda, Z., King, S., & Ibrahim, H. (2017). Is 'summer' reading loss universal? Using ongoing literacy assessment in Malawi to estimate the loss from grade-transition breaks. *Research in Comparative and International Education*, *12*(4), 461–485. https://doi.org/10.1177/1745499917740657

Stark, H. L., Snow, P. C., Eadie, P. A., & Goldfeld, S. R. (2016). Language and reading instruction in early years' classrooms: The knowledge and self-rated ability of Australian teachers. *Annals of Dyslexia, 66*(1), 28–54. https://doi.org/10.1007/s11881-015-0112-0

Suggate, S. P. (2016). A meta-analysis of the long-term effects of phonemic awareness, phonics, fluency, and reading comprehension interventions. *Journal of Learning Disabilities*, *49*(1), 77–96. https://doi.org/10.1177/0022219414528540

Tiruchittampalam, S., Nicholson, T., Levin, J. R., & Ferron, J. M. (2018). The effects of preliteracy knowledge, schooling, and summer vacation on literacy acquisition. *The Journal of Educational Research, 111*(1), 28–42. https://doi.org/10.1080/00220671.2016.1190911

Tunmer, W. E., Chapman, J. W., Greaney, K. T., Prochnow, J. E., & Arrow, A. W. (2013). Why the New Zealand National Literacy Strategy has failed and what can be done about it: Evidence from the Progress in International Reading Literacy Study (PIRLS) 2011 and reading recovery monitoring reports. *Australian Journal of Learning Difficulties, 18*(2), 139–180. https://doi.org/10.1080/19404158.2013.842134

Tunmer, W. E., Greaney, K. T., & Prochnow, J. E. (2015). Pedagogical constructivism in New Zealand literacy education: A flawed approach to teaching reading. In W. E. Tunmer & J. W. Chapman (Eds.), *Excellence and Equity in Literacy Education: The Case of New Zealand* (pp. 121–144). https://doi.org/10.1057/9781137415578_6

Turner, L., & Tse, L. (2015). Solving summer slide: Strategies and suggestions. *Set: Research Information for Teachers (Wellington),* 1, 16–24.

Welch, B., & Bennett, B. (1963). *Summer holiday*. On Summer Holiday.

Williamson, R., Jesson, R., & Shepherd, D. (2020). The summer learning journey: Ameliorating the summer learning effect using blogging. *Computers and Education*. https://doi.org/10.1016/j.compedu.2019.103738. 145.

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